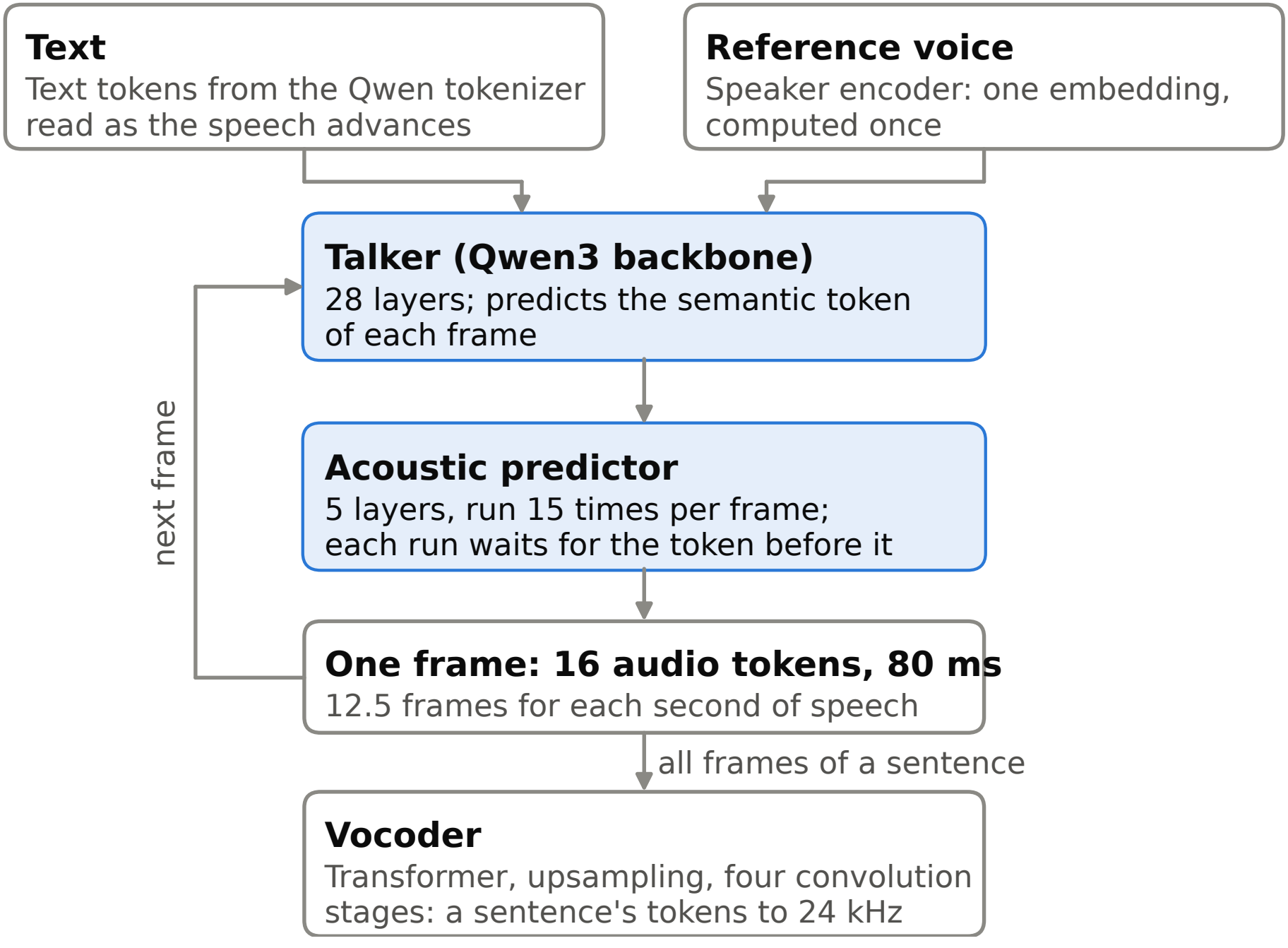
Text
Text tokens from the Qwen tokenizer read as the speech advances
Reference voice
Speaker encoder: one embedding, computed once
Talker (Qwen3 backbone)
28 layers; predicts the semantic token of each frame
Acoustic predictor
5 layers, run 15 times per frame; each run waits for the token before it
One frame: 16 audio tokens, 80 ms
12.5 frames for each second of speech
next frame
all frames of a sentence
Vocoder
Transformer, upsampling, four convolution stages: a sentence's tokens to 24 kHz